

Comparative analysis of chaotic maps in different phases of Crested Porcupine Optimizer

Introduction

Crested Porcupine Optimizer (CPO) is a metaheuristic algorithm that contains 4 defense stages i.e., sight, sound, odor, attack. (sight and sound = exploration, odor and attack = exploitation)

Initially a porcupine population is defined these porcupine population are the solutions. In the original CPO random numbers are used for initializing population of porcupines

But the problem with original CPO is that it might get stuck in local optimum and also premature convergence. To solve this problem multiple enhancements are done to original CPO. One of them is using chaotic maps. Chaotic maps create a sequence of numbers across a range that makes the sequence more uniform spread, deterministic, doesn't repeat values, and sensitive to initial values.

Chaotic maps are based on dimensions like 1D, 2D,...nD. But most research studies use mostly 1D and rarely research goes beyond 3D or 4D. Some examples of 1D and 2D chaotic maps are :

1D = Circle, logistic, tent, sine, chebyshev, piecewise, etc

2D = Henon, lozi, ikeda, cat, baker, etc

Most of the studies that use chaotic maps in CPO often use only 1 chaotic map (like sine, logistic, sinusoidal piecewise map, logistic-tent hybrid map) for population initialization. Along with chaotic maps mutation strategies (like gaussian, cauchy, levy flights), adaptive parameter control, etc are also used which helped in solving the problems encountered in original CPO.

Research Gap

- Multiple studies are done on Chaotic Crested Porcupine Optimizer (CCPO) but most of them use only 1 chaotic map. There is 1 research that compares 10 different chaotic maps and then selects 1 best map.
- Chaotic maps are used only for population initialization and not at any of the 4 defense stages

The objective of this research is

- To compare population initialization with 1 chaotic map and hybrid population initialization (combine 3 different maps)
 - Then compare every individual chaotic map performance using the metrics mean fitness, best fitness, convergence speed, standard deviation, success rate and determine the best chaotic map based on statistical test Friedman, Wilcoxon
 - Initialize population using only 1 best chaotic map
 - Initialize population using top 3 best chaotic map
- And also to evaluate whether chaotic maps integration in 4 defense stages of CPO improves the performance as compared to chaotic population initialization
 - (i) Compare chaos added at sight and sound stage (exploration) with chaos added at odor and attack (exploitation)
 - (ii) Measure mean fitness, best fitness, convergence speed, standard deviation of each method (chaos at initialization, chaos at exploration, chaos at exploitation)
 - (iii) make conclusions based on statistical test Friedman, Wilcoxon

References

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